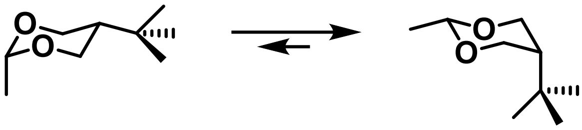
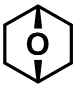
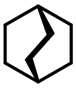
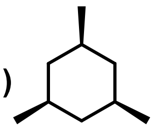
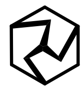
CHM102 Basic Organic Chemistry

Conformations

Q1. Cycloahexanes with bulky substituents in the equatorial position rather than the axial position are more stable. However, the following equilibrium shows the preference for bulky substituent in the axial position. What could be the reason?

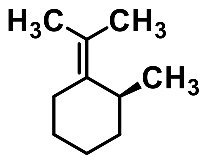


Q2. Draw the proper conformer of the following compounds.

(a)  (b)  (c)  (d)  (e) 

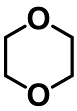
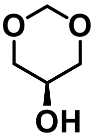
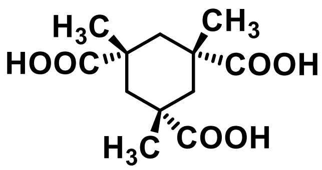
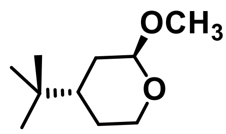
$C_{10}H_{16}$

Q3. Draw the possible chair conformers of the following compounds. Which one of the chair forms is stable? Indicate the reason.

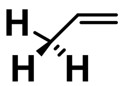


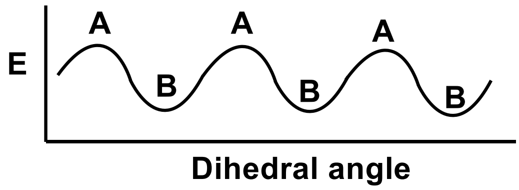
Q4. Indicate the possible conformers of hydrazine (NH_2NH_2), (fluoromethyl)amine (FCH_2NH_2) and fluoromethanol (FCH_2OH) using Newman projection and indicate the stability order with reasons.

Q5. Draw the most stable conformations for the following compounds:

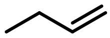
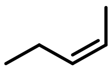
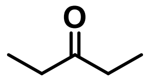
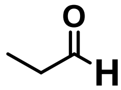
(a)  (b)  (c)  (d) 

Q6. If the conformational changes can be represented for propene as follows, identify the conformations A and B and indicate the reason.





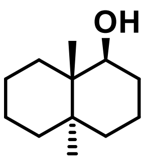
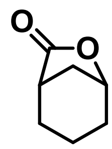
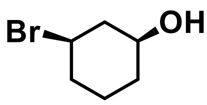
Q7. Identify the major conformers and indicate the most preferred conformations of the following compounds with reason:

(a)  (b)  (c)  (d) 

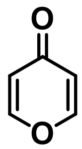
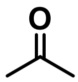
Q8. Identify the conformational preferences in 1,3-butadiene and 3-buten-2-one.

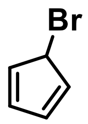
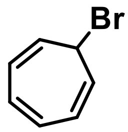
Q9. Draw the Newman projections of planar and puckered cyclobutane and compare the possible strains in each one of them.

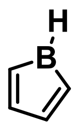
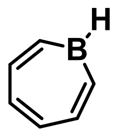
Q10. Draw the conformations of the following compounds.

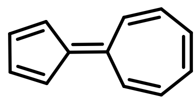
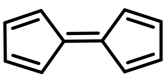
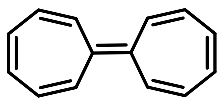
(a)  (b)  (c) 

Q11. Identify the molecule among each pair exhibiting the dominant property indicated for each case and indicate the reason.

(a) Basicity  vs 

(b) Ionization of the C-Br bond  vs 

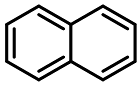
(c) Lewis acidity  vs 

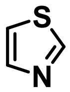
(d) Dipole moment   

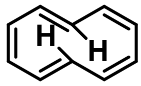
Q12. Draw the molecular orbitals (MO) diagrams of the following with appropriate energetic ordering:

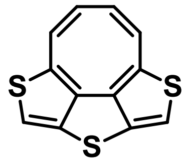
(a) Cyclopentadienyl cation
(b) Cyclopentadienyl radical
(c) Cyclopentadienyl anion

Q13. Classify the following molecules according to their aromaticity characteristics:

(a) 

(b) 

(c) 

(d) 

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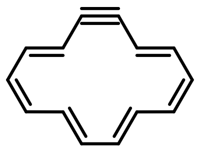
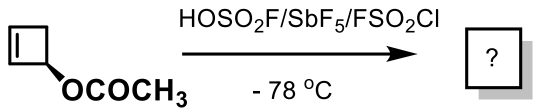
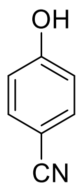
(e) 

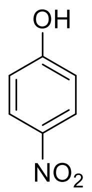
(f) 

Aromaticity of this species

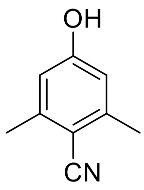
Q14. The pK_a of 1 and 3 are nearly the same, but 2 has lower pKa value than that of 4. Why?



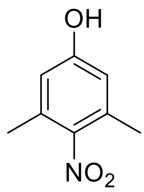
1



2



3



4

Q15. Explain why phenol is more acidic (pKa = 10) than cyclohexanol (pKa = 16).

Q16. Predict the possible dimeric interaction mode in CH_3Cl. (Hint: Use the bond dipole(s)/molecular dipole moment for prediction of the charge centres)

Q17. Nitromethane (pKa = 10) is more acidic than methane (pKa = 48). Why?